2-BIT BINARY ADDER — TRUTH TABLE

Inputs: A = A1 A0 and B = B1 B0 | Output: S = S2 S1 S0

All Possible Input and Output Combinations

#	A1	A0	B1	B0	A	B	A+B	S2	S1	S0
1	0	0	0	0	0	0	0	0	0	0
2	0	0	0	1	0	1	1	0	0	1
3	0	0	1	0	0	2	2	0	1	0
4	0	0	1	1	0	3	3	0	1	1
5	0	1	0	0	1	0	1	0	0	1
6	0	1	0	1	1	1	2	0	1	0
7	0	1	1	0	1	2	3	0	1	1
8	0	1	1	1	1	3	4	1	0	0
9	1	0	0	0	2	0	2	0	1	0
10	1	0	0	1	2	1	3	0	1	1
11	1	0	1	0	2	2	4	1	0	0
12	1	0	1	1	2	3	5	1	0	1
13	1	1	0	0	3	0	3	0	1	1
14	1	1	0	1	3	1	4	1	0	0
15	1	1	1	0	3	2	5	1	0	1
16	1	1	1	1	3	3	6	1	1	0

Example: 01 + 11 = 100, therefore 1 + 3 = 4. The maximum possible result is 11 + 11 = 110 (3 + 3 = 6).

Bit positions

A1/B1: Most Significant Bits (MSB) **A0/B0:** Least Significant Bits (LSB) **S2:** Final carry / most significant result bit.

A 2-bit adder combines two 2-bit binary numbers. Since 3 + 3 = 6 = 110■, the output requires 3 bits: S2 S1 S0.